

LIMBURG REGL, Belgium

WMO: 064700

Lat: 50.792N

Lon: 5.202E

Elev: 246

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.1	18.6	8.1	8.4	14.6	13.7	11.1	20.7	31.4	47.9	28.5	44.4	7.2	50	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	83.0	68.7	80.0	67.4	76.8	65.6	70.3	80.4	68.5	78.1	66.7	75.0	7.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	98.7	75.9	64.8	92.8	73.6	63.2	87.4	71.2	34.4	81.3	32.8	78.6	31.4	75.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	21.4	18.9	9.7	89.9	11.5	4.1	1.4	92.8	-5.3	95.2	-11.8	97.5	-20.1	100.5		
(5)			9.1	74.3	11.3	3.1	1.0	76.5	-5.6	78.3	-12.0	80.0	-20.2	82.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	35.9	35.7	42.5	46.8	54.4	59.0	63.7	63.0	57.6	52.3	42.8	39.6	(6)
(7)		DBStd	11.81	9.25	8.73	6.16	5.95	6.57	6.11	5.62	4.95	5.38	5.91	7.08	6.76	(7)
(8)		HDD50	1848	438	402	243	130	28	3	0	0	3	43	231	327	(8)
(9)		HDD65	5768	902	820	698	545	332	199	95	99	229	396	665	788	(9)
(10)		CDD50	1681	1	2	10	35	166	272	426	404	232	113	16	4	(10)
(11)		CDD65	126	0	0	0	0	4	19	56	38	8	1	0	0	(11)
(12)		CDH74	1325	0	0	0	0	85	198	558	400	77	7	0	0	(12)
(13)		CDH80	281	0	0	0	0	9	24	141	99	8	0	0	0	(13)
(14)	Wind	WSAvg	8.3	10.9	9.6	9.7	8.3	7.1	6.8	6.5	6.5	7.2	8.3	8.8	9.7	(14)
(15)	Precipitation	PrecAvg	30.9	2.6	2.3	2.2	1.9	2.4	2.9	3.0	3.2	2.3	2.5	2.6	2.9	(15)
(16)		PrecMax	40.6	4.9	5.6	5.3	4.6	5.5	6.6	9.0	8.5	6.9	6.9	4.3	6.1	(16)
(17)		PrecMin	23.6	0.1	0.2	0.1	0.2	0.9	0.6	1.2	0.6	0.6	1.2	0.2	0.7	(17)
(18)		PrecStd	4.6	1.2	1.2	1.1	1.0	1.1	1.3	1.6	2.0	1.5	1.1	1.0	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	57.8	64.6	71.4	80.2	83.0	88.3	88.2	81.1	74.2	61.2	56.0	(19)
(20)			MCWB	51.5	51.3	54.5	57.1	63.6	68.2	72.0	69.5	68.7	64.2	54.0	52.0	(20)
(21)		2%	DB	50.8	53.5	58.8	66.4	76.1	79.5	83.9	82.3	75.7	67.9	57.8	53.6	(21)
(22)			MCWB	48.0	49.5	51.6	54.7	61.9	66.5	70.2	68.0	65.9	60.3	53.1	50.4	(22)
(23)		5%	DB	49.0	49.9	54.3	61.4	72.4	75.8	80.5	78.2	71.2	64.0	55.2	51.0	(23)
(24)			MCWB	46.8	46.9	49.6	51.7	60.4	64.4	68.5	66.9	63.2	57.9	51.9	48.6	(24)
(25)		10%	DB	47.2	47.1	51.3	58.1	68.1	71.4	76.4	74.5	67.3	60.8	52.8	48.9	(25)
(26)			MCWB	45.1	44.4	47.3	50.3	58.6	62.8	66.3	64.9	61.2	56.6	50.4	46.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	52.6	55.5	58.8	66.3	70.8	73.8	71.0	69.7	64.4	55.9	52.9	(27)
(28)			MCDWB	53.5	56.5	62.7	69.5	78.2	80.0	84.5	81.9	78.5	72.4	59.0	54.6	(28)
(29)		2%	WB	48.8	49.7	52.6	55.7	63.5	68.2	71.4	69.1	66.7	61.1	54.0	50.7	(29)
(30)			MCDWB	50.1	52.9	57.5	64.3	73.4	76.9	82.1	79.9	74.8	66.1	56.8	53.0	(30)
(31)		5%	WB	47.1	47.3	50.1	53.2	61.0	65.3	69.2	67.6	64.0	59.0	52.2	48.8	(31)
(32)			MCDWB	48.6	49.7	53.6	59.9	70.5	73.5	78.4	77.1	69.3	63.2	54.8	50.8	(32)
(33)		10%	WB	45.0	44.8	48.3	51.1	58.8	62.9	67.0	65.6	62.1	57.0	50.4	47.1	(33)
(34)			MCDWB	47.0	47.1	50.8	57.2	67.2	70.1	75.0	73.4	66.4	60.5	52.8	48.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	11.4	12.5	16.1	17.4	16.3	18.5	15.7	13.7	10.7	8.2	(35)	
(36)			MCDBR	9.4	14.6	16.8	23.1	25.5	23.8	24.9	25.6	23.4	18.1	13.6	8.9	(36)
(37)			MCWBR	8.7	11.7	11.7	13.6	13.9	12.8	13.5	13.4	13.8	11.4	10.1	7.7	(37)
(38)		5% WB	MCDWB	9.2	14.0	15.0	21.2	22.5	21.9	23.1	23.5	19.5	16.8	12.0	8.6	(38)
(39)			MCWBR	8.9	11.5	10.9	13.0	13.3	12.7	13.4	13.2	12.1	11.1	9.4	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.322	0.345	0.378	0.411	0.421	0.419	0.423	0.415	0.386	0.361	0.336	0.320	(40)	
(41)		taud	2.409	2.368	2.282	2.210	2.227	2.249	2.239	2.286	2.354	2.418	2.412	2.376	(41)	
(42)		Ebn at Noon	223	247	259	262	265	266	263	259	254	238	214	202	(42)	
(43)		Edh at Noon	20	26	34	41	42	42	41	38	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	245	458	825	1313	1563	1701	1611	1372	1032	609	289	192	(44)	
(45)		RadStd	34	90	117	149	150	150	193	140	106	50	33	29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page